

TABLE OF CONTENTS

I-200 PLANT MANAGEMENT SYSTEM

	Page No.
1. GENERAL	I 200-2
2. CODES AND STANDARDS	I 200-2
3. DESIGN REQUIREMENTS	I 200-2
4. FUNCTIONAL REQUIREMENTS	I 200-3
4.1 General	I 200-4
4.2 Security and Availability of the Plant Management System	I 200-4
4.3 Function to be Performed	I 200-4
4.4 Detailed Function	I 200-5
4.4.1 Scanning and Alarming	I 200-5
4.4.2 Data Logging	I 200-10
4.4.3 Plant Start-up Sequence Monitor	I 200-11
4.4.4 Performance Calculation	I 200-13
4.4.5 Graphic Display	I 200-14
4.4.6 Historian	I 200-15
4.4.7 Plant Maintenance Log	I 200-16
4.5 Interface with Other Systems	I 200-16
4.6 System Hardware	I 200-16
5. MAL OPERATION	I 200-16
6. POWER FAILURE	I 200-16
7. EQUIPMENT FAILURE	I 200-17
8. SYSTEM SHUT-DOWN	I 200-17
9. CLOCK SYSTEM	I 200-17
10. ACCESSORIES	I 200-17
11. SELF DOCUMENTATION CAPABILITY	I 200-18
12. TESTING	I 200-19
13. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	I 200-19

I 200 PLANT MANAGEMENT SYSTEM

1. GENERAL

The Plant Management System shall be provided by the Contractor to carry out supervisory monitoring, data and event logging, batch processing and performance calculations and maintenance logging.

The Plant Management System is to be designed to improve long term aspects of operation such as security, availability and economy, as well as short-term aspects such as alarming and guidance during incidences when plant limits and constraints are approached.

The Plant Management System is to provide a vital service during the commissioning of the GTG, HRSG, STG, and other items of plant in this project, and shall be available for this duty.

The computer system shall be suitable for operating in the environment at Site as specified elsewhere in this Specification.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards.

3. DESIGN REQUIREMENTS

Instrumentation and control system furnished by the Contractor shall provide the control and/or monitoring functions for the gas and oil fired combined cycle power plant equipment as described in Part 4 Technical Requirements as applicable.

This Specification describes the functional requirement for the Plant Management System. The system hardware and software to meet the requirements of this Specification shall be furnished in accordance with Part 4, Technical Specification I-005, "Distributed Control System".

The general electric requirements governing the design and supply of monitoring equipment shall be as detailed in Part 4, Technical Specification I-001, "Instrumentation and Controls".

The Plant Management system shall also communicate with the other digital control systems. The information shall be transmitted via communication data links such as the plant bus.

The contractor shall provide additional indicators, recorders, transmitters, alarm points, etc. which may be added as a result of review during the detailed design to meet the intent of the Specification.

The hardware shall consist of dual data servers forming a redundant central data historian for the short and long term storage, archiving and retrieval of power station information. There shall be sufficient storage in this system for at least 12 months of historical data for the plant. Data concentration techniques may be used to achieve this.

The contractor shall provide additional indicators, recorders, transmitters, alarm points, etc. which may be added as a result of review during the detailed design to meet the intent of the Specification.

Operator terminals and the engineering terminals shall all have direct access to the historical data for displaying trends and event logs.

The Plant Management System shall be able to retrieve information from the database using multiple search criteria, including plant area, time & date, variable limits. The system shall have links to external plant-wide information system, with appropriate security against data corruption, unauthorised access, and spread of computer viruses.

4. FUNCTIONAL REQUIREMENTS

The plant management systems shall have the necessary software for plant performance and efficiency calculations to supervise and optimize the power station's operation and to present complete overview of the plant's condition.

The plant management system shall provide high value business information to the plant management. The result of the decisions made by the management shall be taken directly back to the Plant Automation systems.

The plant management system shall present the technical and physical relation in the form of curves, bar graphs, characteristics and selectable tabular text of the current fault or alarm in chronological order.

4.1 General

The Plant Management System shall be designed that the operator, by means of his functional keyboards, is able to call up for printing on logging devices, record on the recorders, or display on either VDU all data relating to:

- The associated plant
- The General Services Auxiliaries plant
- Plant electrical and Substation

Communications between plant operators and the computer system shall be based on the use of clear English language text messages and tag numbers, and shall in general conform to the identification systems used elsewhere in the plant and in the control system.

Displays similar to those on the DCS shall be available. See Part 4 Section 3 I-005 “Distributed Control System”

The Plant Management System shall be designed to ensure that no single fault shall cause non-availability to the same standards of availability as set out in Part 4 Section 3 I-005 “Distributed Control System”

4.2 Security and Availability of the Plant Management System

Security and availability of the Plant Management System shall be the same as for the DCS. See Part 4 Section 3 I-005 “Distributed Control System”

4.3 Function to be Performed

The Plant Monitoring System shall perform the following functions:

- Process status reporting
- Sequence of event reporting
- Performance calculations
- Operator control functions
- Operator guidance functions
- Data storage and management reports
- Maintenance log report

4.4 Detailed Function

The detailed function of Plant Management System shall be in accordance with the following

4.4.1 Scanning and Alarming

The detailed function of Plant Management System shall be in accordance with the function of the scanning and display functions of the DCS.

a) Data Acquisition

The Control System shall accept, scan, monitor and process all the types of signal inputs specified and shall provide the required signal outputs, as specified, consistent with the readout devices, final elements etc.

The Control System shall perform all system support calculations, special calculations, automatic calculations as required in the routine handling of all input data such as averaging, linearisation, curve fitting, integrating etc, and shall generate the calculated data base. The Control System shall have the capability to detect alarms and changes of state and shall be able to transmit this information throughout the system, checking each point in the system against its established limit.

The Control System shall be able to perform self-diagnostics and provide error and alarm condition at the operator interface, thereby advising the operator of imminent system failure and prescribing corrective actions.

The Contractor shall be responsible for the preparation of all programs including diagnostics required for a debugged and completely operational system. The Contractor shall develop the system software in a user-friendly and modular form so that changes can be isolated without impacting the entire system, and can be readily implemented without complexity of programs or prototypical programs. The system shall be designed to allow the programmer to make online changes, accessing the system in a conversational mode through an interactive interface.

All equations developed shall be documented with their respective derivations. All computations shall be referenced as to their source and shall be identified with their source point(s) and intermediate point(s) tag name identification.

b) Scan and Alarm

(i) Scanning

The system shall continually and automatically scan, limit check and update all analogue, digital, and pulse inputs from the plant used in monitoring and control functions.

(ii) Alarm Priority Levels

The VDU alarming function shall permit the assignment of each analogue and digital alarm condition to any of four priority levels. This feature shall display the alarm condition such that an operator can easily identify the priority level that has been assigned. It shall be possible to change the alarm priority level assignments of points through the engineering console or the Operator console. Note that the consoles shall have password protection, enabling only limited access. Each priority level shall have a flashing message and a different colour.

The alarms shall system shall enable the operator to automatically navigate from the alarm, through the dispatch hierarchy, to the control area affected, showing the control tile in discrepancy.

One audible tone shall be given when any alarm is activated. Alarm suppression shall be available for all priority levels. Alarm suppression (alarm cut-outs) means preventing alarms from being reported when not a genuine alarm condition such as preventing a low-pressure alarm when a pump is turned off.

All alarms shall be capable of being displayed on the VDU by a sectable filter. The alarm summary will be display new (flashing 2Hz) and pending (acknowledged) alarms, until the alarm condition is cleared (the cleared alarm will be automatically

cleared from the display). Only alarms will be shown on the alarm summary screen.

Alarm messages shall include:

Priority code

Description

Tag reference number

Time tag

Individual or queued alarms shall be acknowledged from the operator work station or local control panel.

All alarm points shall be time tagged at the time of occurrence. Time tagging of the alarm at the display time will not be acceptable.

(iii) Analogue Alarms

All analogue inputs and composed points (see v below) shall be compared against high and/or low alarm setpoint each time they are measured. The setpoints will be fixed values as manually established or computed functions of other variables. When a measurement is determined as being outside of pre-set limits, the time, the point identification (including its alphanumeric description) current value of the point and its alarm limit shall be displayed on the VDU. The time shall be in hour/minute/second.

Alphanumeric descriptions for inputs shall consist of 30 characters. The alphanumeric characters shall be defined as described by the Purchaser.

A similar message shall be displayed when the alarm returns to normal.

All analogue inputs and composed points with alarm limits shall be assigned an individual “dead band” value. Before an analogue input, or composed point in the alarm state is defined as returned to normal, the analogue input shall have returned to

the non-alarm operating region a distance equal to or greater than the dead band.

Variable alarm setpoints shall be possible for all inputs. It is intended that for these points the alarm setpoint will vary as a function of another variable such as load, flow and temperature. The variable alarms shall be able to be assigned to all points in the data base.

It shall be possible to inhibit the alarming on each of the analogue inputs, calculated variables and composed points by the logical state of one or more binary input points or the values of analogue variables in order to eliminate nuisance alarms. The ability of inhibiting binary inputs or analogue value shall be different for each analogue input or composed point if required. When inhibited, no alarming indication shall be given to the operator. It shall also be removed from any alarm summaries. Upon restoration of alarm functions, if the point is in alarm, the operator shall be informed by the usual means (current status is given).

(iv) Bad Input Alarm (channel fault)

Each analogue input, calculated variable and composed point shall be checked against process limits (in the cases of thermocouples, the thermocouples will also be checked to determine if the thermocouple is open circuit. Each binary/analogue I/O shall be provided with the ability to detect out of range/fault.

If the quality of the analogue input or composed point is “bad”, a message shall be displayed on the VDU screen informing the operator that the input is bad. This quality shall include “out of scan” status.

(v) Alarming of Composed Points

Provisions shall be made to limit-check composed points. These will be assigned point identification numbers, alphanumeric descriptions, and alarm limits and will be treated as analogue

inputs. Composed points are defined as (1) the difference between two inputs, (2) the sum, average, minimum or maximum value of several inputs, (3) the rate of change of an input, or (4) rate of change of an averaged group of inputs or composed points.

(vi) Sequence of Event (SOE)

The system shall provide scanning of up to 50 digital (contact) inputs for Sequential Event Recording. These inputs shall be scanned to discriminate between contact operations that occur a minimum of five milliseconds apart and print them in their proper sequence when they are opening and closing. Each sequence of events shall be displayed as an individual sequence.

The time shall be displayed in hours, minutes, seconds and milliseconds with each sequential event contact status change.

A sequence of events is considered complete when all digital sequence of events (SOE) inputs have been sensed or a period of time as designated by the Purchaser has elapsed since the last detected status change. The system shall be capable of recording and storing up to two separate sequence of events logs without data loss. When the event is completed, the operator shall be able to initiate a print out of the sequence of events log on the logging printer from the operator work station. The program shall operate to record the events chronologically and shall send the sequence of events log in its entirety.

Contractor shall provide sufficient buffer storage in the data acquisition system memory structure in the SOE programs to ensure that the system will not fail to detect or record a contact operation due to storage filling.

4.4.2 Data Logging

- a. The Plant Management System shall be capable to provide the following recording functions or logs:
 - (1) Alarm log
 - (2) Trend log
 - (3) Post trip log
 - (4) Sequence of events log
 - (5) Hourly and daily log
 - (6) System review

The Contractor shall provide details on log selection techniques, and shall indicate the respective function keys employed for printing/logging selection.

- b. Trend Log

The Plant Management System shall have provisions for manual initiation from the operator console for individual or group data logging of plant status in accordance with the trends in the DCS:

- c. Post Trip Log

In the event any trip should arise on the plant or main equipment of the plant, the data at the predetermined 256 analog points contained in sufficient number of group shall be printed out at intervals of 10 seconds starting from 5 minutes before occurrence of such trip and 10 minutes after such trip.

- d. Sequence of Events Log

In the event any trouble occur in the plant or major plant items including generator circuit breaker the Plant Monitoring System shall be capable to print out automatically all related binary data/input points status in time sequence order on the alarm or data printer with the time resolution not higher than 1 millisecond. Those binary input may be put in the groups.

e. Hourly and Daily Logs

The Plant Management System shall have provision for automatic and manual initiation from the operator console to print preselected analog and binary inputs in the data printer:

f. System Review

The Plant Management System shall allow, on demand by the operator, the following information to be printed out by the data printer or displayed on the VDU:

- (1) Alarm summary (binary and analog points)
- (2) Bad input summary
- (3) Deleted input summary
- (4) Alarm lockout summary
- (5) Simulated points summary
- (6) Log of post trip trigger
- (7) Plant start-up and shut down guidance

4.4.3 Plant Start-up Sequence Monitor

Various automatic sequences can be initiated by the operator or automatically. Each of these sequences shall have a sequence monitoring display associated with it. The design of the screen shall assist the operator in following the progress of the sequence.

As a minimum, that display is to display a list of all pre-start permissives and all sequence steps with an indication of whether or not a permissive is satisfied or a sequence step has been completed.

In addition, if any sequence step fails to complete in a time determined by the sequence controller, the sequence monitor shall raise an alarm and display it on the alarm display, irrespective of whether the sequence monitoring screen is in use or not. This and the following shall apply to both manually initiated and automatically initiated sequences.

- Ability to disable sequence block (switch to manual).

- When sequence function block executed (start-up or shutdown) the indication should be different between start-up/shutdown in-progress to start-up/shutdown complete (e.g. blinking when in-progress and steady state when complete).
- Ability to logically condition release, auto command, protection and complete criteria on sequence function block.

Each sequence step is :

- ⇒ Independently monitored and displayed to the Operator interface (reflecting status of each step).
- ⇒ Correct Step Bypass criteria, allows sequencer to synchronise with the process to minimise on unnecessary repeats in operations (e.g. if the fan is in service then there is no need to close the isolating damper again).
- ⇒ Navigational tool to allow operator to determine missing criteria (can be in dynamic text format or logical expression showing each step criteria).
- ⇒ When a step goes out in monitoring time, its output must be terminated and the operator can manually rectify the situation, and the sequence will automatically continue once the discrepancy has been acknowledged.
- ⇒ Process criteria used as feedback complete rather than step output.
- ⇒ Sequence block will initialise all bias and setpoint stations to default value, as well as placing related controllers to auto (unless bypassed by maintain selection – setpoint or controller).

Every operator station shall have its own processor (CPU). Hence, if one fails, another station shall take over the function of the failed one and the operation shall be maintained without any disturbance.

Each Operator Station shall be loaded from its own bulk memory automatically when power failure is recovered.

Operator Stations shall share printers and auxiliary storage devices through Ethernet LAN.

Access to the Operator Station shall be provided by a multi-level password system.

A dedicated auxiliary storage device shall be provided on each Operator Station. The peripheral of HMI auxiliary devices and printers shall be market available device to ease the long-term maintenance and cost of ownership.

4.4.4 Performance Calculation

Performance calculations shall provide reliable information to the performance of the plant. These calculations shall assist in enabling attainment of optimum plant operation efficiency and performance.

For this purpose, the actual plant operating condition shall be compared with targets of the performance which shall be based upon the unit load and other variable parameters. The above target values of the performance shall be calculated based upon the boggle curve memorized in the computer.

The calculation formulas for the efficiency of gas turbine, HRSG and steam turbine plant shall, in principle, be the same formulas as specified in “Guarantee and Performance” which are used for the performance test. The formulas of Input/output method shall be applied.

The efficiency and performance shall be calculated for the following items.

- (1) Gas Turbine efficiency
- (2) HRSG efficiency
- (3) Steam Turbine efficiency
- (4) Total Plant efficiency at Generator end
- (5) Total Plant efficiency at Sending out end
- (6) Condenser performance.
- (7) Exhaust emission and pollution level calculation
- (8) And other equipment/systems as suggested by the Contractor

The Contractor shall submit the detailed formulas and methods for each calculation to the Owner for approval. These formulas and items to be subjected by performance calculation shall be included in the Bid proposal.

4.4.5 Graphic Display

The Contractor shall be responsible for providing the following preprogrammed graphic displays. These graphic displays shall also be used for operation of the plant. See Part 4 Section 3 I-005 “DCS”.

- a. *Plant start-up trend curves for cold, warm and hot start-up, consisting of:*
 - Gas turbine speed curve
 - Steam turbine speed curve
 - Gas turbine load curve
 - Steam turbine load curve
- b. *Process and Instrumentation Diagrams (P&IDs)*
 - HP and LP steam diagram
 - HP and LP Bypass system diagram
 - HP and LP section of HRSG system
 - Feed water exhaust gas system
 - Condensate system
 - Air and exhaust gas system
 - Gas/Oil fuel system
 - Circulating Water system
 - Auxiliary plant cooling system
 - Gas Turbine and Steam turbine lube oil system
 - Steam Turbine drain, vent and sealing system
 - Condenser tube cleaning system
 - Auxiliary Steam system
- c. *Plant Electrical Diagrams*
 - 380 V plant electrical and emergency diesel system single line diagram
 - 6 kV electrical single line diagram

- 500 kV Switchyard Extension single line diagram

d. Condition Supervisory Information

- Generator and excitation and generator cooling system
- Bearing temperature of all major equipment
- HRSG metal temperature monitoring diagram

e. X-Y Plots

The system shall have the capability to display a real time point-plot of a variable Y as a function of another variable X. These variables shall be assigned as X and Y through the operator's consoles. It shall also be possible to plot up to four variables as a function of one variable.

In particular, X-Y plots shall be developed for the following:

- generator capability diagram
- head flow curves of major pumps and fans.

4.4.6 Historian

Facilities shall be available to archive all the data received in the DCS and Plant Management System.

The central data historian shall collect and store the data locally for a minimum of twelve months without the need for storage media changes.

The system shall have the facility to copy records to long term storage media for archiving (e.g. on CD). A suitable method of indexing, access and retrieval of data once copied to permanent storage shall be provided via menu functions.

All digital and analogue signals (measurement, control and actuation signals) from all the systems, as well as all events, warnings, alarms and trips need to be logged in real time. The time at which the event etc. is tagged shall be the time of the event itself. Time tagging at the time the signal is received at the operator interface is not acceptable.

All data shall be available in a form suitable for use in Microsoft formats and shall be able to be stored for the life of the power station.

4.4.7 Plant Maintenance Log

The system shall monitor parameters which affect major plant equipment maintenance scheduling and shall provide a calculated prediction of routine maintenance.

The maintenance log shall also advise what replacement parts are necessary for the predicted routine maintenance and identify any such parts which are not in inventory.

4.5 Interface with Other Systems

The system shall be a sub-system of the DCS. As such, it shall have access to all the data received in the DCS including hard-wired I/O, remote I/O and data received from links to PLCs.

4.6 System Hardware

Refer to Part 4, Technical Specification I-005 “Distributed Control System”.

5. MAL OPERATION

The system shall be provided with adequate precautions against accidental or deliberate mal operation. Access to vital parts of the software shall be limited by password system and/or key lock.

It shall not be possible to corrupt any program or any operating system data from an Operator Station.

Suitable notices indicating restricted access shall be placed on any door giving access to the Electronic Equipment Room or to terminals with unrestricted facilities.

6. POWER FAILURE

As a sub-system of the DCS, this system shall exhibit the same behaviour as the DCS on loss of power. See Part 4 Section 3 I-005 “DCS”

7. EQUIPMENT FAILURE

Any malfunction within the computer system shall not be permitted to inhibit the correct functioning of the plant.

Interlocking shall be provided to ensure that failure in one location shall not cause damage elsewhere.

8. SYSTEM SHUT-DOWN

In the event of the system being shut-down for any reason, sufficient information shall be preserved in non-volatile store to permit an orderly computer restart. This information shall not be liable to corruption during shutdown, whether due to power failure or other causes.

9. CLOCK SYSTEM

The Contractor shall provide a master and slave clock system for the power plant with the signal originating from a satellite communication device.

The master clock shall be located in the central control room and shall operate from the 220 V DC system or from a 220 V AC UPS system.

All monitoring system internal clocks shall be connected and synchronized to this clock. The master clock shall operate the slave dials located in every room in the central control building.

10. ACCESSORIES

Adequate and appropriate accessories shall be provided by the Contractor. These are to include, but not be limited to, the following items:

- (1) Desks, with chairs, for computer operators with adequate writing space, and including storage facilities for reference documents.
- (2) Sufficient shelving, racks, cupboards or other storage to accommodate all operating, maintenance, and other system reference manuals, together with storage for the specified stationery supplies, and for spare parts.
- (3) Secure storage (fire-proof, heat-proof, and magnetically shielded as appropriate) for data storage media.

- (4) Filing facilities for printed output.
- (5) Recording equipment for temperature and humidity for each electronics equipment room.
- (6) Any other accessories which are appropriate to the specific system or installation.
- (7) All necessary cabling (permanent and test), terminals and racks, power supplies, and so on shall be provided with the system.

11. SELF DOCUMENTATION CAPABILITY

The Management System shall have the capability of generating self documentation to simplify the drawings and other document control during design, check out, commissioning and ultimate Owner's reconfiguration during the plant life.

The Contractor shall furnished an integrated documentation for the system which shall be modular in design. The Document shall be divided into major areas such as System Document, User's Document and Engineering Document. The system documents shall discuss what the system does, providing the Owner with useful information on the system's configuration and specification, and an overview of the system. The User's documentation shall provide the information on how the system is used and other important information for operator and system engineering. The Engineering documents shall provide information on how the system is built and works and other detailed engineering information.

The Contractor shall provide all the forms and detailed procedures for engineering and configuration of the Plant Management System. This shall include methods for defining system functions, equipment partitioning, cabinet arrangements, power supply auctioneering, system and loop configuration procedures for analog controllers, on/off and sequential and interlock logic, monitoring functions, output functions, and assigning functions to various modules.

The self documentation capability shall be provided through Engineering Station even though the system modifications shall be possible through the Engineering Station and the test panel. The documentation shall be generated in worksheets and flow chart forms.

The Contractor shall provide complete software documentation as described in Part 4, Subsection 2.2.

The Contractor shall provide a complete software configuration management program in order to identify the configuration of the Plant Management System at discrete points in time for the purpose of controlling the development of the system and maintaining the integrity and traceability of the configuration throughout the system life cycle.

12. TESTING

Refer to Part 4, Technical specification I-005 “Distributed Control System”.

13. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Contractor shall conform to the quality assurance/quality control requirement of Part 4, Subsection 2.7 Clause 4 unless the mechanical equipment with which the instrumentation and controls are associated is level 1 in which case Clause 3 shall be applicable.

(Last page of I-200)